

JK LIFESTYLE DIGITAL ECOSYSTEM

Technical Architecture, Execution Roadmap & Costing Plan

Website | Super App | Enterprise Platform (ERP) | Clinical | Data, Events & AI

Prepared for	JK Lifestyle Ltd. (Group Management Office)
Prepared as	CTO / Technical Architect board proposal and execution baseline
Baseline	JK Lifestyle Technical Development Plan, September 2026
Planning horizon	24 months
Budget currency	BDT (Bangladeshi Taka)

Budget status: this is a planning baseline, not a vendor quotation. Final approval should follow Human Resources salary validation, cloud sizing, license choices and supplier quotations.

1. Executive Summary

Architectural baseline. The program uses a staged transition instead of making Odoo Community Edition the permanent Point of Sale (POS) and Inventory platform. First, the team cleans the existing Excel product, outlet and opening-stock data and imports it into Odoo Community Edition so POS and Inventory can go live quickly. In parallel, JK Tech builds the long-term Enterprise Resource Planning (ERP), Inventory and POS modules. The custom ERP manages clinical workflows, employee and Human Resources workflows, payroll and all management and operational reports. The JK-built POS and Inventory replace Odoo outlet by outlet only after parallel testing and reconciliation are complete. JK ID, the central integration gateway, security controls, business events and the ERP reporting layer remain common across the ecosystem.

Metric	Base Planning Value	Comment
Program duration	24 months	Overlapping phase delivery; two-week sprint cadence
Build-team peak	~14 active technical team members at peak	The team grows by phase. No separate Product Owner, Data/AI, AI-model or Community Moderator position is included in the current baseline.
24-month development & specialist cost	BDT 26.9M	Loaded in-house cost plus outsourced Odoo/UIUX support; consolidated roles absorbed by Tech Lead
Cloud / platform run cost	BDT 10.0M	Gradually increases with App, media and AI usage
Other delivery costs	BDT 11.5M	AI/API, SaaS, security, equipment, migration/training, recruitment
Base spend before contingency	BDT 48.4M	Cash plan for approved work
Contingency reserve	BDT 5.8M (12%)	Held centrally; released only through change/risk control
Total planning envelope	BDT 54.3M (~5.43 crore)	Recommended approval ceiling for the 24-month baseline

Important exclusions. Payment and mobile-financial-service transaction charges are business transaction costs tied to sales volume and are not included in the project envelope. The baseline uses Odoo Community Edition for the initial POS and Inventory transition, so no Odoo Enterprise license is included. Any paid Odoo hosting, third-party add-ons, support contracts, taxes/VAT, or unusually high video/AI consumption should be budgeted separately if required.

2. Planning Assumptions & Guardrails

- Target design scale: 11 outlets initially, 500-700 daily patients, 5,000 peak concurrent users and 50,000+ registered online customers in Year 1. Infrastructure is sized progressively, not pre-purchased for full scale.
- JK Tech is treated as having no pre-existing engineering bench at project start; recruitment is therefore on the critical path.
- Odoo Community Edition is a temporary transition system for outlet Point of Sale (POS) and Inventory. Existing product, outlet and stock data will be cleaned from Excel and imported into Odoo Community Edition. JK Tech will build the long-term Inventory and POS modules in parallel and replace Odoo outlet by outlet after user acceptance testing and reconciliation. The custom Enterprise Resource Planning (ERP) system will generate all management and operational reports throughout the transition.
- No big-bang migration will be used. Excel-to-Odoo migration and later Odoo-to-JK POS/Inventory migration will use staged rehearsal, parallel operation, reconciliation, rollback planning and outlet-by-outlet cut-over.
- Two-week sprints are mandatory. Every sprint ends with a demo and targeted user acceptance testing; each major go-live requires regression, performance, security and business acceptance gates.

- Identity, the custom ERP/Clinical platform and platform operations remain in-house. Odoo specialist work may be outsourced with knowledge transfer. Product planning, data and artificial-intelligence work, machine-learning oversight and community-moderation governance are covered by the Tech Lead / Architect in this budget instead of separate full-time roles.
- Cloud and third-party budgets are reviewed monthly with alerts at 70%, 85% and 100% of the approved envelope.

3. System Architecture & Technical Design

Architecture objective. Build one coherent digital operating platform with explicit ownership boundaries, a shared identity/control plane, API-first integration and event capture from day one. The design prioritizes business continuity, maintainability, health-data protection and future service extraction without paying day-one microservice operational cost.

3.1 Architecture decisions & guardrails

Decision	Choice	Technical rationale
Architecture style	Modular monolith / bounded domains	5-6 domain-aligned modules/services; extract only when scaling, compliance or workload isolation requires it.
ERP strategy	Staged transition: Odoo CE first, JK-built POS/Inventory long term	Use Odoo Community Edition to stabilize outlet POS and Inventory quickly after Excel migration. Build the long-term JK Inventory and POS in parallel, then replace Odoo only after the new modules pass parallel testing and reconciliation. The custom ERP remains the long-term clinical, Human Resources, payroll and reporting platform.
Cloud posture	Hybrid-leaning-cloud	Managed cloud for applications/data, with local resilience for clinical/outlet continuity where internet dependence is unacceptable.
Integration model	API-first + domain events	No direct cross-domain database access; synchronous APIs for commands/queries, events for decoupled downstream reactions.
Identity	One JK ID	A single party identity can hold Customer, Patient, Member, Employee or Vendor roles without duplicate customer records.
Migration	Parallel run, no big bang	Rehearse in staging, shadow/parallel run, reconcile, preserve rollback and cut over outlet/module by module.
Delivery	Continuous user acceptance testing + gated releases	Two-week sprints, demos, targeted user acceptance testing, security/performance/disaster-recovery evidence before major go-live.

3.2 Layered system architecture

Layer	Primary components	Responsibility
Experience / Channel	Super Website; Super App; Staff Apps; JK ERP/Clinical Console; Odoo CE POS during transition	These are the customer, workforce, clinical and outlet interfaces. Odoo CE supports the first POS/Inventory phase, while JK ERP is the long-term operational and reporting platform.
Shared Foundation / Control Plane	JK ID and single sign-on; central integration gateway; role-based access control; consent; notifications; content management; search; payment; loyalty; audit	This layer provides one identity, central access rules, common communications and shared controls across every application.
Business Domains	POS/Inventory transition; Clinical/Care; Employee and Human Resources; Membership/Fitness; Community/Education; ERP Reporting	Odoo Community Edition is temporary for POS/Inventory. JK-owned modules manage clinical services, Human Resources, reporting and the long-term POS/Inventory replacement with clear data ownership.
Data, Events & Intelligence	PostgreSQL; Redis; RabbitMQ; Object Storage / Content Delivery Network; Warehouse; AI and machine learning services	Operational persistence, cache/session, asynchronous events, media, analytics and model workloads.
Integration	Payment/mobile financial services; SMS/Email/WhatsApp; Health Connect/HealthKit; wearable system interfaces/Bluetooth; Sinbad; biometric/lab devices	Adapters isolate external providers and Odoo integration contracts from custom domain logic, preserving replaceability and controlled coupling.

- Boundary rule: each domain owns its data; other domains consume APIs or events, never shared tables.
- Evolution rule: Clinical and AI/Food Recognition are the first likely candidates for independent service extraction because of health-data isolation and inference workloads.
- Data rule: events required for alerts, analytics and future prediction are captured from day one; historical data cannot be retrofitted later.
- Migration rule: no production cutover without rehearsal, reconciliation, rollback and business-owner sign-off.

3.3 Domain ownership & ERP boundary

Domain	System of record / ownership	Scope
Identity & Shared Platform	Custom / JK Tech	JK ID, login, role-based access control, consent, notifications, audit logs, the central integration gateway, integration rules and identity mapping to Odoo customer/partner records.
Initial Outlet POS & Inventory	Transition: Odoo CE → JK Tech	Odoo Community Edition initially runs outlet POS and Inventory using cleaned Excel data. JK Tech builds the long-term Inventory and POS modules and takes ownership after

Domain	System of record / ownership	Scope
		a controlled outlet-by-outlet cut-over. Accounting is treated as a separate integration decision and is not the reason for keeping Odoo as the long-term POS or Inventory platform.
Clinical and Care Plan	Custom / JK Tech with clinical expert support	The ERP manages doctor profiles, patient registration, appointment requests, outpatient services, patient medical records, consent, Care Plans, clinical invoices, tasks and alerts.
Employee and Human Resources Management	Custom / JK Tech	The ERP manages employee records, attendance, leave, roster, payroll calculation and payroll approval.
Digital Commerce Experience & Booking	Custom channels + active sales backend	The Website and Super App own the customer experience. The sales adapter sends orders to Odoo during transition and to the JK-built Inventory/POS platform after cut-over.
Membership / Fitness / Unified Customer View	Custom / JK Tech	Membership lifecycle, classes, profile and Customer 360 read model over shared JK ID and cross-system references.
Community / Education / Franchise	Custom / JK Tech	Community feed, moderation console, learning/video, franchise/distribution and field-operation workflows.
Analytics and Artificial Intelligence	Custom / JK Tech	The data and artificial-intelligence layer supports analytics, the data warehouse and future predictive models. All management and operational reports are presented through the ERP Reporting module.
Central Reporting	Custom ERP / JK Tech	The ERP generates all management and operational reports. During the Odoo phase it reads synchronized Odoo data; after cut-over it reads the JK-built POS/Inventory directly.

ERP and POS/Inventory transition boundary. Odoo Community Edition is the first operational bridge for outlet POS and Inventory, not the final long-term platform for those modules. Existing Excel product, outlet and opening-stock data is cleaned and migrated into Odoo CE. At the same time, JK Tech builds the custom ERP, central Reporting, Inventory and POS modules. The new Inventory/POS will run beside Odoo during pilot testing, and outlets will move only after sales, payments, returns, stock movements and closing balances reconcile. After cut-over, the JK Inventory/POS becomes the operational source of truth. The ERP remains the single reporting interface before, during and after migration.

3.4 Identity & data architecture

Entity	Purpose	Primary consumers
party (JK ID)	Identity root: name, contact, authentication identity and consent linkage	All channels and custom domains; mapped to Odoo partner/customer reference
party_role	Customer / Patient / Member / Employee / Vendor role linkage	Unified Customer View, Clinical, Membership and Human Resources
odoo_product / lot / stock reference	Migration mapping between legacy Excel records, Odoo CE records and later JK Inventory records	ERP reporting, Website/App catalogue, migration reconciliation
appointment	Patient, doctor, request/status, schedule, location and service context	Custom ERP/Clinical, App/Website booking, notifications
care_plan / target / task	Plan versions, typed targets and daily/weekly tasks	Clinical authoring, App planner, alerts
device_reading	Time-series glucose, weight, blood pressure, steps, sleep, source and timestamp	App, Alert Engine, Warehouse
clinical_bill	Clinical services, professional/service context, approval status and Odoo invoice reference	Custom Clinical workflow; Odoo financial posting/payment
attendance / payroll_run	Attendance, leave/roster and payroll calculation with approval and reporting status	ERP Human Resources/Payroll and central Reporting
commercial_order reference	Sales/order reference that points to the active backend: Odoo during transition and JK POS/Inventory after cut-over	Website/App, ERP Reporting, support and analytics
reporting_event	Normalized sales, stock, online-order, clinical, attendance and payroll events used by the ERP reporting layer	ERP reports, dashboards, management summaries and analytics

- Product and stock traceability must remain continuous across the transition. The ERP keeps stable mapping between legacy Excel IDs, Odoo CE records and later JK Inventory IDs so historical sales, batches/lots and reports do not break when an outlet moves.
- Care Plan extensibility: targets use metric_code + value + unit rather than fixed columns, so new clinical metrics do not require repeated schema migrations.
- Customer 360 is a read model over the shared JK ID and role relationships, not a second master customer table.

3.5 Integration, offline sync & real-time event architecture

Area	Pattern	Control / behavior
Offline POS	Odoo CE offline/reconnect validation → later JK POS offline store + synchronization → reconciliation	Offline behavior is tested in both phases. The JK POS must safely replay transactions and prevent duplicate sales/payments after reconnect.
Stock conflict	Reconciliation during Odoo phase; JK Inventory becomes authoritative after cut-over	Odoo stock is authoritative during the transition. The JK Inventory runs in parallel for testing. After a successful outlet cut-over, the JK Inventory becomes authoritative for that outlet. Duplicate submissions use transaction IDs and reconciliation reports.
Cash/shift close	Odoo POS first; JK POS later	During transition, outlet cash/session closing is handled by Odoo POS. The JK POS must match the same sales, payment and closing totals before an outlet is moved.
Wearables / continuous glucose monitor	Health Connect + HealthKit first	Use a direct vendor system interface or Bluetooth connection where required. Manual entry always remains available through the same backend service.
Event backbone	RabbitMQ first	Single ingestion path can feed Alert Engine, notifications and Warehouse; Kafka only if proven event volume justifies it.
Notifications	FCM/APNs + WebSocket	Domain events trigger mobile push; WebSocket supports outpatient queue and live manager dashboards.
Reliability	Idempotency + retry + trace IDs	Repeat-safe processing, correlation IDs and observable retry behavior for offline and vendor-failure scenarios.
Community feed	Hybrid consistency	Immediate consistency for a user's own write; ranked feed re-scored asynchronously.
Clinical alert flow. Dangerous reading → Alert Engine rule → patient notification and clinician task in parallel → clinician response-time rule → escalation to on-call supervisor when unacknowledged. The platform provides reliable delivery and an audit trail; it is monitoring infrastructure, not an emergency-response replacement.		

3.6 Security, privacy & resilience architecture

Control area	Implementation
Authorization	Use role-based access control by business concern, location and job function. Access is enforced by the API/service layer, not only by the screen.
Authentication	Staff and administrator users use two-factor authentication. Customers and patients use a

Control area	Implementation
	one-time-password login to keep onboarding simple.
Encryption	TLS in transit, encryption at rest, plus field-level encryption for sensitive patient medical records/device/Care Plan fields.
Audit	Sensitive health-data access logged on reads as well as writes; correlation/trace IDs retained for investigations.
Consent	Explicit, revocable consent records govern device sharing, community use and analytics/AI processing.
Application security	Run dependency scanning, secure-code checks and standard web-security controls in the automated build pipeline. Use independent security testing before major go-lives and at least annually afterward.
Backup / disaster recovery	Use automated backups, restore tests and module-specific recovery-time and acceptable-data-loss targets. Complete a disaster-recovery rehearsal before major public releases.
Local resilience	During the Odoo phase, validate offline and reconnect behavior for outlet POS. The later JK POS includes offline transaction capture and safe background synchronization.
Observability	Use structured logs, request trace IDs, performance metrics, agreed service performance targets, security-issue tracking and monthly cloud-cost review.

3.7 Technology platform & DevOps

Layer	Choice	Reason
Web	Next.js (React) + TypeScript	Server-side rendering improves search visibility, and the shared component approach keeps the website/admin UI consistent.
Mobile	Flutter	One Android/iOS codebase, with offline storage and Bluetooth Low Energy device support.
Backend	Node.js / NestJS	Strong modular boundaries, TypeScript contracts and API-first service design.
AI / data services	Python / FastAPI	machine-learning/data-science ecosystem and model-serving path.
ERP	Odoo Community Edition for the transition + custom JK ERP for long-term operations	Odoo Community Edition provides the first POS/Inventory bridge. JK ERP owns clinical services, Human Resources/payroll and all reports, and JK Tech later replaces Odoo POS/Inventory with in-house modules.
Primary database	Managed PostgreSQL	Managed PostgreSQL provides reliable transactions for finance and clinical data, with flexible JSON fields where needed.
Cache/session	Redis	Session, rate limiting, cache and selected queue/state use cases.
Events	RabbitMQ → Kafka only if justified	Operationally simpler at current scale; future throughput path remains

Layer	Choice	Reason
		open.
Search	PostgreSQL full-text search → OpenSearch when needed	Start with PostgreSQL search. Add OpenSearch only when catalogue, community or course volume requires it.
Object/media	S3-compatible object storage + Content Delivery Network	Store food/progress images and education media with compression, Content Delivery Network delivery and lifecycle policies.
Runtime	Managed Kubernetes	Dev/Staging/Prod parity, horizontal scaling and workload isolation.
automated build/deployment	GitHub Actions + Argo CD (or equivalent)	Every production change is traceable to source control and passes a controlled deployment process.
Reporting	ERP reporting layer + PostgreSQL reporting/read models	All management and operational reports are generated from the ERP. Data is synchronized from Odoo during transition and from JK-owned modules after cut-over.

3.8 Data & AI architecture

Pipeline step	Architecture	Governance / control
1. Capture	Food photo, user correction, device reading, adherence outcome	Write consent/data-lineage metadata with the event.
2. Recognize	Third-party food-recognition API + locally curated Bangladeshi nutrition data	Buy phase accelerates launch while local data remains JK-owned.
3. Human confirm	Confidence score + correction when below threshold	Manual correction is a first-class product path and safety control.
4. Store & label	Consented events land in warehouse/training dataset	Corrections become labeled examples; preserve source and model/version lineage.
5. Build decision	Local model proof of concept after sufficient labeled data exists	Compare quality, latency and cost before shifting inference in-house.
- Indicative Phase-2 quality target: $\geq 80\%$ correct dish identification with an approximate $\pm 20\%$ macro/portion estimate target; always expose confidence and confirmation when uncertain. - Inference cost control: cache common/repeated meals and batch non-urgent recognition rather than paying synchronous vendor cost for every photo. - Warehouse prerequisites: event capture, stable metric_code conventions, consent tagging and data lineage are established before predictive models depend on them.		

3.9 Key end-to-end system flows

Flow	Path	Architecture intent
Care Plan	ERP Clinical Screen → Clinical Service → PostgreSQL → business event → App integration → patient	Clinical remains the authoring source of truth. The App receives the Care Plan through a controlled system interface instead of reading clinical database tables directly.
Meal / AI	Super App → object storage → food-recognition service/local nutrition data → user confirmation → business	Separate media, inference and analytics paths; corrections

	event → data warehouse/adherence calculation → clinician dashboard	become labeled data with consent and lineage.
Online / e-commerce sale	Website/Super App → Sales Adapter → Odoo CE during transition → JK Inventory/POS after cut-over → payment/fulfillment → ERP Reporting	The customer experience does not change when the backend changes. The adapter points to the active sales/inventory backend, and all results are reported through the ERP.
Outlet / POS sale	Excel data → Odoo CE POS/Inventory first → parallel JK POS/Inventory build → shadow reconciliation → outlet-by-outlet JK cut-over → ERP Reporting	Odoo CE gets outlets running quickly. JK Tech then replaces POS and Inventory after the new system proves that sales, payment and stock totals match.
Clinical invoice	Custom ERP/Clinical → clinical service selection → invoice approval → ERP clinical invoice/payment status → ERP Reporting	The custom ERP owns the complete clinical invoice workflow, including the service context, invoice approval and payment status. If a separate accounting system is required, the approved financial summary can be exported or integrated without making Odoo the owner of the clinical invoice.
Payroll	Custom ERP attendance and payroll → payroll approval → ERP payroll records → ERP Reporting → optional accounting export	The custom ERP owns attendance, payroll calculation, payroll approval and payroll reporting. If the Finance team needs accounting entries in another accounting system, approved payroll totals can be exported or integrated separately.
Offline POS	Odoo CE offline/reconnect validation → later JK POS offline store and background synchronization → reconciliation	Offline behavior is tested in both phases. The JK POS must safely replay queued transactions and prevent duplicate sales or payments after reconnect.
Authentication	Website/App/Staff channel → JK ID and single sign-on → role and consent check → secure session → central integration gateway → custom module or Odoo connector	Identity and authorization remain centralized while Odoo customer/employee references are mapped, not duplicated as a competing identity master.
Central reporting	POS/Inventory + Online Sales + Clinical + Human Resources/Payroll → ERP Reporting Layer → Management Reports	The ERP is the single report interface. During transition it consumes Odoo data; after cut-over it consumes the JK-built operational modules directly.

3.10 Deployment topology & non-functional baseline

Area	Baseline design	Control / target
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Environment model	Separate Development, Staging and Production environments with isolated passwords, keys and configuration	Every deployment traceable to source control; production promotion is gated.
Cloud runtime	Managed Kubernetes + managed PostgreSQL + Redis + RabbitMQ + object storage and content delivery	Scale progressively rather than provisioning Phase-5 capacity on day one.
Connectivity	Cloud-first, with protected network access for clinical operations and tested outlet offline capability	Clinical and outlet operations should continue during internet problems and synchronize safely after connectivity returns.
Scale design	11 outlets initially; 500-700 daily patients; 5,000 peak concurrent users; 50,000+ Year-1 registered users	Horizontal scaling, caching and content delivery reduce pressure on the main application servers as usage grows.
Resilience	Automated backups, restore tests, defined recovery-time and acceptable-data-loss targets and disaster-recovery rehearsal	Clinical and financial data receive stricter recovery targets than community and general content workloads.
Observability	Structured logs, performance metrics, request trace IDs, service performance dashboard and alerting	Review reliability, security issues and cloud cost every month using agreed thresholds.

3.11 Concise answers to key board questions

These concise answers summarize the controls already defined in the architecture, delivery, risk and operating-cost sections.

Q	Question focus	Concise answer
Q10	Top project-failure risks	The three biggest risks are: (1) dirty Excel/Odoo/master data, controlled by early data cleaning, staged migration and reconciliation; (2) changing clinical workflows, controlled by early prototypes and clinical expert approval; and (3) team capacity or scope pressure, controlled by phased hiring, clear acceptance criteria, formal change control and quarterly delivery gates.
Q14	Community moderation at scale	Use a hybrid moderation model. Technology automatically flags possible health misinformation, and trained people make the final moderation decision through the Moderation Console within a defined review time. The Community feature cannot go live until the moderation tools and minimum human review capacity are ready.
Q15	Responsibility for dangerous alerts	A dangerous reading triggers the Alert Engine. The patient is notified and the assigned clinician receives a task at the same time. The clinician must respond within a defined time; if there

is no response, the alert is escalated to the on-call supervisor. Queued events, safe retries, delivery monitoring, request IDs and audit history make the process reliable. The platform supports monitoring but is not an emergency-response service.

Q16

Controlling media and AI run cost

Right-size computing capacity with automatic scaling limits. Compress images, use content-delivery caching, use adaptive video quality and move old media to cheaper storage. Reuse common meal-recognition results and batch non-urgent artificial-intelligence processing. Track storage, content-delivery and artificial-intelligence unit costs every month, and move from third-party services to an in-house model only when quality and economics justify it.

4. 24-Month Delivery Roadmap

Scheduling model. The phases intentionally overlap. Dependencies are respected (identity/master data before integration testing; Clinical authoring before the App Care Plan loop), while independent teams work in parallel to reduce calendar time. User acceptance testing happens throughout the program rather than as a final-stage activity.

4.1 Months 1-12

Month	Primary Focus	Major Deliverables	Acceptance / user acceptance testing Gate
M1	Mobilize & discover	Confirm business owners; create the architecture decision log; review Excel, Odoo and current systems; define security/performance requirements; prioritize the backlog.	Scope v1 approved; risk register ready; backlog prioritized.
M2	Foundation + data preparation	Set up Development, Staging and Production environments, automated deployment, the central integration gateway, JK ID, master-data rules and detailed Excel-to-Odoo migration mapping.	Environment ready; identity approved; Excel data-quality report completed.
M3	Excel to Odoo CE migration	Clean product, outlet and opening-stock data from Excel; configure Odoo CE Inventory and POS; create migration scripts and ERP reporting synchronization.	Migration rehearsal #1; opening stock reconciled; pilot dataset approved.
M4	Odoo CE outlet pilot + ERP start	Run the first Odoo CE outlet pilot for POS and Inventory. Start Doctor Profile, Appointment and patient medical-record design. Define the first ERP report catalogue.	Outlet user acceptance test passed; stock and sales reconciled; clinical workflow approved.
M5	ERP workforce + reporting core	Keep Odoo CE live for outlet POS and Inventory. Build employee management, attendance, leave and roster, payroll, patient registration, appointment/outpatient queue and the first ERP sales, stock and clinical reports.	Outlet workflow accepted; employee workflow accepted; first report reconciliation passed.

Month	Primary Focus	Major Deliverables	Acceptance / user acceptance testing Gate
M6	Odoo transition Gate 1	Stabilize the Odoo CE pilot, reconcile sales, payments and stock, train outlet users and approve requirements for the JK-built Inventory and POS modules.	Gate 1: Odoo CE pilot accepted; reconciliation passed; independent security test #1 completed.
M7	Odoo rollout + JK Inventory build	Move the first outlet wave to Odoo CE. Build the JK Inventory service, continue patient medical records and consent/audit work, and connect the website through a sales adapter that can later switch away from Odoo.	Outlet wave accepted; Inventory prototype reviewed; medical-record tests approved.
M8	JK POS build + App starts	Build the JK POS core on the new Inventory service. Continue Care Plan authoring, website sales integration, Flutter foundation, notifications and food-recognition service procurement.	JK POS alpha accepted; Website beta reviewed; mobile architecture approved.
M9	Parallel POS/Inventory test	Run Odoo CE and the JK-built POS/Inventory side by side at a pilot outlet. Compare sales, returns, payments, stock and closing balances while continuing the Clinical and App pilot.	Parallel reconciliation passed; clinical pilot accepted; website regression passed.
M10	Clinical + Website v1 and first JK cut-over	Go live with Doctor, Patient, Appointment, outpatient workflow, patient medical records and basic Care Plan. If reconciliation succeeds, move the first approved outlet from Odoo CE to the JK POS/Inventory modules.	Gate 2: production go-live accepted; first outlet cut-over accepted; independent security test #2 completed.
M11	Care loop + outlet migration	Continue App planner, food log, reminders, membership and customer-view work while moving additional approved outlets to the JK POS/Inventory platform.	App features accepted; outlet reconciliation remains clean.
M12	AI, devices + migration review	Start food-recognition service use, Health Connect/HealthKit integration, moderation tools and analytics events. Review the remaining Odoo-to-JK outlet migration plan.	AI quality baseline agreed; migration progress reviewed.

4.2 Months 13-24

Month	Primary Focus	Major Deliverables	Acceptance / user acceptance testing Gate
M13	Super App beta + JK POS/Inventory rollout	Run the Super App beta and continue outlet migration to the JK POS/Inventory modules with sales, stock and payment reconciliation.	Closed beta; outlet cut-over evidence approved.
M14	Super App v1 go-live	Release the end-to-end Care Plan loop, Food Log AI and basic wearable path while keeping central ERP reporting across all modules.	Gate 3: App v1 accepted; independent security test #3.
M15	Complete main POS/Inventory migration	Move the remaining priority outlets from Odoo CE to the JK POS/Inventory platform and close major operational gaps.	Priority outlet migration accepted; Odoo transition reduced.
M16	Community live + reporting scale	Release Community/Education v1 and expand ERP reporting and central data pipelines.	Content-safety gate; report reconciliation passed.
M17	Franchise & staff operations	Add franchise/distribution, selected partner integrations and field staff applications.	Integration acceptance; field pilot.
M18	Phase 4 Gate	Complete Community/Education/Franchise release, operating procedures, training and ownership handover.	Gate 4: business acceptance; operational handover.
M19	Reporting and analytics scale	Expand ERP dashboards for outlet sales, Inventory, online sales, clinical operations, attendance and payroll.	ERP reports reconcile with all source systems.
M20	Predictive analytics prototype	Prototype adherence/drop-off analysis, review alert quality and strengthen dataset governance.	Offline model evaluation; clinical expert review.
M21	In-house food model proof of concept	Train on the local dataset, test model serving and compare build cost with third-party API cost.	Model quality/cost decision gate.
M22	Scale-out	Support 15-20 outlets, tune performance, test autoscaling and complete a disaster-recovery rehearsal.	Load/performance acceptance; disaster-recovery test passed.
M23	Hardening & transition	Complete security hardening, audit cleanup, upgrade rehearsal, documentation and knowledge transfer.	Knowledge transfer accepted; runbooks reviewed; resilience tested.
M24	Program close / steady operations	Stabilize the platform, close critical backlog items, review the service performance dashboard, confirm the annual operating model and approve the Year-3 roadmap.	Gate 5: normal operational ownership accepted.

5. First 90 Days - Week-by-Week Execution Plan

Week	Focus	Deliverables	Done / Acceptance
W1	Program mobilization	Confirm sponsors, business owners, risks, decision cadence, repository ownership and required system access.	Kickoff minutes + accountable owner per business area.
W2	Current-state & data audit	Review Excel files, current Odoo use, website, outlet operations, clinical workflows, duplicate master data and integrations.	Current-state inventory + migration risk report.
W3	Cloud / engineering baseline	Set up Development, Staging and Production, network access, secrets, repositories, release rules and automated build/deployment.	Environment checklist.
W4	Platform skeleton	Create the central integration gateway, system monitoring, request tracking, authentication design, backup baseline and security standards.	Architecture review #1.
W5	JK ID / single sign-on	Create the person-and-role model, one-time-password login, staff two-factor login, secure session flow, consent linkage and the first end-to-end login.	Identity demo + security acceptance.
W6	Clean and map Excel data	Clean product, outlet, stock, customer and employee data; define mapping IDs; prepare staged imports into Odoo CE and the ERP reporting layer.	Migration rehearsal #1.
W7	Configure Odoo CE POS & Inventory	Configure products, outlets, opening stock, POS sessions, users and permissions; confirm the API and reporting synchronization needed by the ERP.	Odoo CE pilot configuration approved.
W8	Odoo CE outlet pilot	Test sale, return, payment, outlet closing, stock movement, receipt/printer and internet-loss/reconnect behavior in Odoo CE.	Odoo CE POS & Inventory alpha demo.
W9	Reconciliation & replacement requirements	Compare Excel opening data with Odoo, test network failure/reconnect, document gaps, and convert real outlet workflows into requirements for the JK-built POS and Inventory modules.	Reconciliation report + approved replacement backlog.
W10	ERP reporting v1	Build the first ERP reports for outlet sales and stock using synchronized Odoo data, and for online sales using the Website/App order source. Reconcile the combined report totals.	ERP report totals match Odoo for outlet/stock data and the Website/App source for online sales; migration rehearsal #2 completed.
W11	Pilot outlet user acceptance	Train outlet users and test device, printer, payment, stock and reporting workflows. Finalize support and rollback steps.	Pilot-readiness checklist.
W12	Odoo CE live pilot & JK build plan	Run the controlled Odoo CE pilot, close critical defects, reconcile sales/stock, demonstrate ERP reports and approve the next-quarter plan for the JK Inventory and POS build.	Go/no-go for wider Odoo transition + approved JK replacement plan.

6. Workstreams, Parallelism & Critical Dependencies

Workstream	Active Window	Scope	Critical Dependency
Foundation / Platform	M1-M24	JK ID, central integration gateway, automated build/deployment, system monitoring, security, notifications and integration rules	Blocks integration testing for every product stream
Odoo CE POS / Inventory Transition	M2-M15 transition	Clean Excel data, configure Odoo CE Inventory/POS, run pilot/rollout, provide transition APIs and support parallel reconciliation.	Excel data quality + Odoo pilot + outlet reconciliation
Custom ERP / Clinical / Human Resources / Reporting	M4-M24	Doctor Profile, Patient, Appointment, outpatient operations, patient medical records, Care Plan, Clinical Billing, employee management, Attendance, Leave/Roster, Payroll and central Reporting.	Clinical experts + identity + reporting rules + security
Website	M4-M10	Group hub and customer commerce experience, Health Revolution booking, unified login/cart. The sales adapter points to Odoo during transition and the JK Inventory platform later.	JK ID + stable sales/inventory adapter + booking APIs
Super App / Care Plan	M8-M14	Customer app, care plan loop, food log, membership, devices, notifications	Clinical authoring model must be stable
Community / Education	M12-M18	Feed, moderation, learning platform, content/video, expert presence	Moderation console and content policy are hard go-live gates
Franchise / Staff Apps	M14-M18	Distribution, corporate/health tourism integrations, field staff apps	Verified partner system interfaces + stable Inventory and identity
Analytics / AI	M1 event capture; M4 Reporting; M16-M24 analytics/AI	Data warehouse, analytics dashboards, adherence analysis and AI model evolution. Business and operational reports remain in the ERP.	Reliable source data + event lineage + consent capture
JK-built POS / Inventory	M7-M16	Build Inventory, stock movement, outlet POS, sale/return/payment, offline/reconnect, reconciliation and phased outlet migration.	Stable business rules from Odoo pilot + clean data + outlet acceptance testing

7. Team Ramp & Responsibility Model

Period	Approx. Team Size	Composition	Primary Delivery Load
M1-2	6.0	Architect, 2 backend, DevOps, 0.5 QA, Odoo specialist, 0.5 UI/UX	Foundation + Excel/Odoo discovery
M3-6	8.5	Architect, 3 backend, DevOps, QA, Frontend, Odoo specialist, 0.5 UI/UX	Odoo CE transition + early ERP/Clinical/Reporting
M7-10	13.5	Architect, 4 backend, DevOps, 2 QA, 2 Frontend, 2 Mobile, Odoo specialist, 0.5 UI/UX	JK Inventory/POS build + Clinical/Website + App start
M11-14	14.0	Architect, 4 backend, DevOps, 2 QA, 2 Frontend, 3 Mobile, ERP/Odoo engineer	Super App + outlet migration + Reporting
M15-18	12.0	Architect, 4 backend, DevOps, 2 QA, Frontend, 2 Mobile, ERP/Odoo engineer	Finish POS/Inventory migration + Community/Franchise
M19-24	11.0	Architect, 3 backend, DevOps, 2 QA, Frontend, 2 Mobile, ERP/Odoo engineer	Reporting, analytics, hardening and knowledge transfer

Role consolidation. The current budget does not include separate positions for Product Owner, Data/AI Engineer, AI Specialist or Community Moderator. The Tech Lead / Architect covers those responsibilities. Clinical subject-matter experts are provided by Health Revolution and are not counted as additional payroll, but their availability is still required for workflow and safety approval.

8. Cost Model

Costing follows the architecture and delivery model above. The current baseline keeps the same role rates, in-house cost loading, Odoo/UIUX specialist support, infrastructure ramp and centrally controlled contingency. The detailed JK-built POS and Inventory scope should be re-estimated after the first two months of data and process discovery. If the new build is larger than the current team can absorb, the quarterly budget should be updated before development expands.

8.1 Costing basis & numeric assumptions

Why numeric assumptions are required. The baseline document intentionally used relative cost weights instead of fixed currency values. For financial planning, this version converts that shape into a BDT baseline using role-rate assumptions. These rates are placeholders for budgeting and must be replaced by Human Resources-approved salary offers and signed supplier quotations before procurement.

8.2 Role-rate assumptions

Role	Mode	Planning Rate (BDT/month)	Scope
Tech Lead / Architect	In-house	200,000	Architecture, security, vendor governance, product/backlog ownership, process mapping, acceptance criteria, Data/AI & machine learning oversight, community moderation governance, final go-live sign-off
Senior Backend Engineer	In-house	100,000	central integration gateway, core platform, domain architecture

Role	Mode	Planning Rate (BDT/month)	Scope
Mid Backend Engineer	In-house	80,000	Custom ERP/Clinical, employee management, Attendance, Payroll, central Reporting, Odoo transition adapters, Membership and Community services
Frontend Engineer	In-house	80,000	Next.js website/admin experiences
Flutter Mobile Engineer	In-house	70,000	Super App, offline sync, device integrations
QA / Test Engineer	In-house	50,000	Manual testing, automation, release testing and user acceptance coordination
DevOps / Platform Engineer	In-house	100,000	Cloud, Kubernetes, automated build/deployment, security baseline and system monitoring
ERP / Odoo Engineer	In-house	70,000	Odoo CE transition support, ERP reporting integration and migration to JK-built POS/Inventory
Odoo Specialist	Outsourced	70,000 / person-month	Initial Excel migration, Odoo CE Inventory/POS setup, pilot support, integration support and knowledge transfer
UI/UX Designer	Contract	80,000 / month	Design system, low-literacy UX, service flows

8.3 Development and specialist cost by delivery block

Period	In-house Payroll / mo (M BDT)	External / mo (M BDT)	Loaded Dev Cost / mo* (M BDT)	Period Total (M BDT)
M1-2	0.505	0.110	0.691	1.382
M3-6	0.690	0.110	0.904	3.614
M7-10	1.040	0.110	1.306	5.224
M11-14	1.180	0.000	1.357	5.428
M15-18	1.030	0.000	1.185	4.738
M19-24	0.950	0.000	1.093	6.555
TOTAL				26.941

***Loaded development cost.** In-house payroll is loaded with 15% for employer overhead, benefits and recruitment friction; external specialist rates are treated as fully loaded. No separate Product Owner, Data/AI Engineer, AI Specialist or Community Moderator position is budgeted; those responsibilities are included under the Tech Lead / Architect baseline. The separate recruitment/onboarding budget covers hiring campaigns, equipment gaps and exceptional onboarding costs.

Cost Category	24-Month Base (M BDT)	Planning Note
Loaded engineering + outsourced specialists	26.9	Core technical delivery team plus Odoo/UIUX specialist support; consolidated roles absorbed by Tech Lead
Cloud / platform infrastructure	10.0	Application servers, managed database, Redis, messaging, object storage/content delivery and backup/disaster recovery; cost grows by phase

Cost Category	24-Month Base (M BDT)	Planning Note
AI / Nutrition API consumption	1.8	Buy-phase food recognition/nutrition baseline; controlled by caching/batching
Communication / Maps / Monitoring / SaaS	2.4	SMS/email/WhatsApp baseline, maps, APM/logging, low-volume tools
Independent security testing	1.8	Three major pre-go-live security assessments at approximately BDT 0.6M each
Engineering/test equipment & field hardware	2.5	Developer/test devices, mobile devices, POS/peripheral lab and pilot equipment
Data migration, training & change management	2.0	Rehearsals, outlet/clinic training, cut-over support and reconciliation
Recruitment & onboarding reserve	1.0	Hiring campaigns, screening, joining/onboarding costs
Subtotal before contingency	48.4	Approved delivery baseline
Contingency reserve (12%)	5.8	Only released against approved risk/change
TOTAL PROGRAM ENVELOPE	54.3	~BDT 5.43 crore over 24 months

8.4 Infrastructure ramp assumption

Period	Monthly Infra (M BDT)	Period Total (M BDT)	What drives cost
M1-3	0.12	0.36	Foundation/dev/test + small production footprint
M4-6	0.20	0.60	Odoo CE outlet pilot + early ERP/Clinical/Website
M7-12	0.30	1.80	Odoo transition + JK POS/Inventory build + Website/Clinical/App beta
M13-18	0.50	3.00	Super App, media, Community/Education and Reporting/data platform
M19-24	0.70	4.20	Higher usage, reporting/analytics/AI and 15-20 outlet support
TOTAL		9.96	Rounded to BDT 10.0M in the program budget

9. Quarterly Budget Release & Delivery Gates

Budget Window	Base Release (M BDT)	Delivery Focus	Release / Acceptance Gate
Q1 - M1-3	4.79	Foundation + Excel cleanup + Odoo CE migration start	Architecture baseline, environments live, identity demonstrated, Excel data-quality/migration rehearsal complete.
Q2 - M4-6	5.06	Odoo CE POS/Inventory pilot + ERP/Clinical/Reporting start	Odoo CE outlet pilot accepted; sales/stock reconciliation passed; independent security test #1.
Q3 - M7-9	6.06	JK Inventory/POS build + Odoo rollout + Clinical/Website/App foundation	JK Inventory/POS prototype and parallel reconciliation ready; Clinical pilot ready; Website beta accepted.
Q4 - M10-12	7.06	First JK POS/Inventory cut-over + Clinical/Website go-live + App build	First outlet cut-over accepted; Clinical/Website production stable; independent security test #2.
Q5 - M13-15	6.88	Super App go-live + Community/Learning platform beta	App v1 accepted + independent security testing #3; closed-loop care functioning
Q6 - M16-18	6.33	Community/Education/Franchise + warehouse	Phase-4 business acceptance; operational training complete
Q7 - M19-21	6.11	ERP reporting/analytics + AI maturity	ERP reports reconcile with source systems; model quality/cost decision completed.
Q8 - M22-24	6.15	Scale, disaster recovery, hardening and handover	Performance, disaster recovery and knowledge transfer accepted; normal operational ownership signed.
Program contingency	5.81	Central reserve; not pre-allocated to quarters	Release only through Steering Committee change/risk approval

Budget-control rule. Quarterly funds should be approved against the next delivery gate; monthly burn is reviewed by the Tech Lead and Finance. A forecast variance above 5% requires a written recovery plan; above 10% requires Steering Committee scope/time/budget approval.

10. Quality, user acceptance testing, Security & Release Strategy

Cadence / Gate	Required Activities	Exit Condition
Every sprint	Run unit/integration tests, API contract checks, targeted user acceptance tests, demos and security/dependency scans.	Critical defects are triaged before the next sprint.
Every month	Regression pack, cloud-cost review, vulnerability backlog, data reconciliation checks	Monthly delivery + burn report
Before outlet/clinic rollout	Migration rehearsal, staff training, parallel run, rollback plan, support roster	Business owner cut-over approval
Before major public go-live	Run performance/load testing, independent security testing, disaster-recovery and backup-restore testing, system-monitoring checks, agreed service performance targets and privacy/consent verification.	The Tech Lead and Business Owner make the formal go/no-go decision.
Post-launch support for 2-4 weeks	Provide focused post-launch support, review incidents, adoption and open defects.	Exit focused support only when agreed service targets and defect levels are stable.

11. Budget & Timeline Risks with Controls

Risk	Exposure	Impact	Control
Hiring delay	High	Pushes critical path because JK Tech starts with no bench	Start Phase-2/3 hiring during Phase 0-1; keep approved specialist bridge contracts
Dirty Excel / Odoo data	High	Bad source data can cause stock errors, failed reconciliation and rollout delay.	Profile and clean data early, stage migrations and perform physical stock checks before outlet cut-over.
Clinical workflow churn	High	Late workflow changes can cause repeated work in patient records and Care Plans.	Keep clinical subject-matter experts involved, prototype early and approve acceptance criteria before full build.
Odoo transition becomes permanent	Medium/High	A temporary bridge can become long-term technical debt if the JK replacement is delayed.	Give the JK Inventory/POS replacement a dated roadmap, run parallel reconciliation and move outlets in controlled waves.
Cloud/media/AI usage spike	Medium	Run-cost exceeds envelope	Compression/lifecycle, caching/batching, quotas, monthly unit-cost dashboard
Vendor/API availability	Medium	Blocks Sinbad, wearable or AI features	Adapter interfaces; manual fallback; defer integrations that are not verified
Scope expansion	High	Uncontrolled additions can increase budget and timeline.	Use a clear Definition of Done, formal change control and quarterly approval gates.
Security/compliance defect	High	A serious defect can delay go-live and create business risk.	Use security-by-design, encryption, access audit logs and independent security testing before major releases.

Risk	Exposure	Impact	Control
Reporting mismatch	High	Management can lose trust if ERP reports do not match outlet, Inventory, clinical or payroll source totals.	Use source references, daily reconciliation and report sign-off during every migration wave.

12. Limited-Budget / Half-Year-1-Budget Fallback

- Defer Community Platform and Education/Learning platform before cutting the shared foundation, Odoo CE POS/Inventory transition, or Custom ERP/Clinical.
- Delay deep Wearable/continuous glucose monitor integration; keep manual entry and basic Health Connect/HealthKit capability where affordable.
- Keep Food Recognition in buy-mode longer and delay the in-house model build; this preserves user value while avoiding early model investment.
- Defer Franchise Management and Health Tourism/Sinbad integration until the core operating system is stable.
- Do not cut JK ID / single sign-on, the central integration gateway, clean master data, core security, the Odoo Community Edition transition needed for current outlet operations, or the custom ERP / Clinical / Reporting foundation.

13. Program Governance & Reporting

Cadence	Meeting / Control	Participants	Output
Daily	Engineering stand-up per workstream	Engineering leads	Blockers, defects, environment issues
Weekly	Architecture/vendor/risk review	Tech Lead + DevOps + module leads	risk/issue/action/decision, architecture decisions, dependency plan, actual progress and spending versus plan
Bi-weekly	Sprint review + user acceptance testing + planning	Business owners + engineering	Accepted increments and next sprint commitment
Monthly	Steering committee	GMO/CEO/CFO/Tech Lead	Budget, timeline, scope changes, risk decisions
Quarterly	Phase gate / budget release	Steering committee	Acceptance gate, next-quarter budget, hiring/vendor decisions

14. Architect Recommendation

Adopt the architecture-led 24-month roadmap with quarterly funding gates and a BDT 54.3M planning ceiling, while treating BDT 48.4M as the delivery baseline and BDT 5.8M as controlled contingency. Keep the transition simple: clean the Excel data, move outlet POS and Inventory into Odoo Community Edition for immediate stability, build the custom ERP and central Reporting, then build and replace POS and Inventory with JK-owned modules after parallel testing and reconciliation. The ERP remains the single reporting interface throughout the program. Scale the Super App, Community and AI only after the operational data is reliable.

- Approve Phase 0 and Q1 budget immediately only after named business owners and data-access owners are confirmed.
- Validate all remaining role rates with Human Resources and replace the Odoo and user-interface-design placeholders with signed quotations before Quarter 2 approval. Validate Odoo hosting/add-on/support costs and third-party artificial-intelligence service costs separately as operating/vendor costs.
- Confirm the Odoo Community Edition version and modules for the transition, validate Excel migration and outlet POS/Inventory operation, and approve the detailed requirements and migration plan for the JK-built POS and Inventory replacement.
- Use business acceptance and measurable technical gates (reconciliation, independent security testing, performance, disaster recovery, user acceptance testing) as the condition for go-live and next-quarter budget release.
- Review the cost model every quarter using actual salary, cloud unit cost, artificial-intelligence cost per food log, SMS cost per active patient, and storage/content-delivery cost per active user.

Appendix A - Budget Formula & Reforecast Method

Cost Driver	Formula	Control
People cost	Sum(role count x monthly rate x active months) + 15% in-house loading	Reforecast on hiring date/rate changes; no separate PO/Data-AI and machine learning/moderator team members in baseline
Cloud cost	Application servers + database + cache/messaging + object storage/content delivery + backup/disaster recovery	Track actual monthly unit cost; forecast from active users/content volume
AI cost	Recognition API calls x unit price + nutrition data + model serving compute	Track cost per confirmed food-log and cache hit rate
Communication	SMS/WhatsApp/email message volume x unit rate	Track messages per active customer/patient
Contingency	12% of approved baseline	Not part of team discretionary spend; requires change/risk approval

Final validation checklist. Before approval, Finance and Human Resources should validate salary bands. Procurement should obtain quotations for Odoo Community Edition implementation/hosting/support, artificial-intelligence services, SMS, cloud infrastructure and independent security testing. Business Owners should confirm the Excel migration volumes and POS/Inventory rules. The Odoo pilot must validate sales, payments, stock, returns and reconciliation. ERP report totals must match the source systems. Legal/Compliance should confirm health-data residency and consent requirements. Any material variance should be included in the next quarterly budget review.